

PRE- BOARD EXAMINATION (2024-25)

CLASS – X

TIME: 3 Hrs SUBJECT: MATHEMATICS STANDARD (041) M.M: 80

INSTRUCTIONS:

This paper is divided into 5 sections. Section A contains 18 multiple choice questions and 2 assertion reason type questions of 1 mark each and all questions are compulsory.

Section B contains 5 very short answer type questions of 2 marks each.

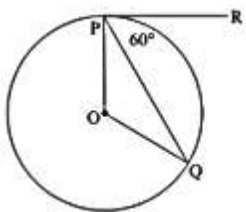
Section C contains 6 short answer type questions of 3 marks each.

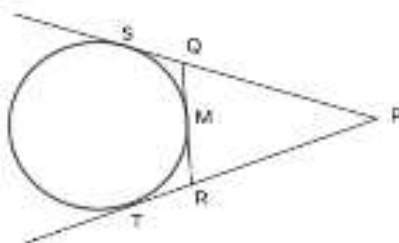
Section D contains 4 long answer type questions of 5 marks each.

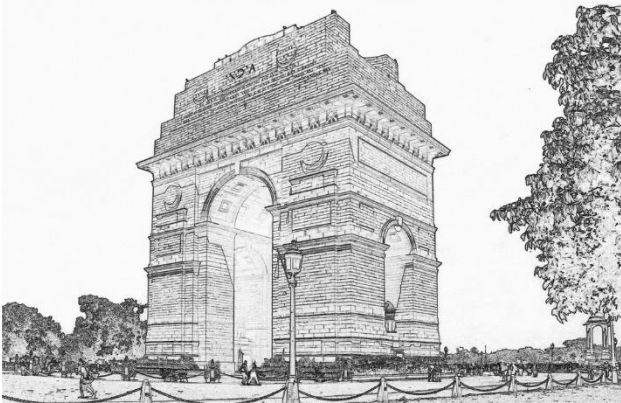
Section D contains 3 CCT type questions of 4 marks each.

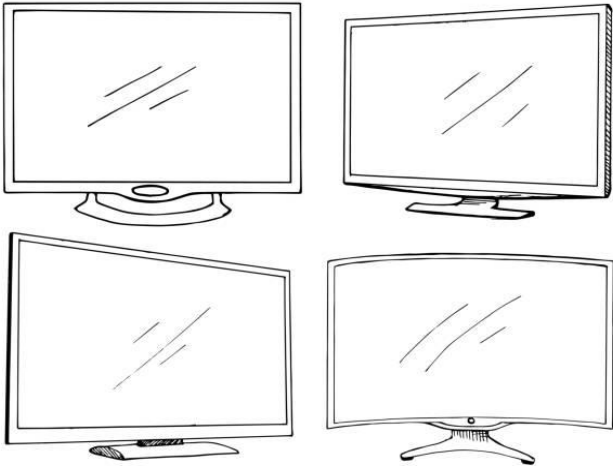
There is an internal choice in 2 questions of section B, 2 questions of section C and 2 questions of section D. Student has to attempt only one in such questions.

S.No.	QUESTION	MARKS
SECTION A [1X20 =20]		
1	Which are the zeroes of $p(x) = 6x^2 - 7x - 3$ (a) 5, -2 (b) -5, 2 (c) -5, -2 (d) none of these	1
2	The ratio in which the X-axis divides the line segment joining A (3, 6) and B (12, -3) is: (a) 2:1 (b) 1:2 (c) -2:1 (d) 1:-2	1
3	The nth term of an A.P. 5, 2, -1, -4, -7 ... is (a) $2n + 5$ (b) $2n - 5$ (c) $8 - 3n$ (d) $3n - 8$	1
4	The length of an arc of a circle with radius 12 cm is 10π cm. The angle subtended by the arc at the centre of the circle, is (a) 120 deg (b) 60 deg (c) 75 deg (d) 150 deg	1
5	If one zero of the quadratic equation $x^2 + 3x + k$ is 2, then the value of k is: (a) 10 (b) -10 (c) 5 (d) -5	1
6	If triangles ABC and DEF are similar and AB=4 cm, DE=6 cm, EF=9 cm and FD=12 cm, the perimeter of triangle ABC is: (a) 22 cm (b) 20 cm (c) 21 cm (d) 18 cm	1
7	The ratio of LCM and HCF of 2 and 4 is (a). 2:4 (b) 2:1 (c) 2:2 (d) 3:4	1
8	In ΔABC right angled at B, if $\cot C = \sqrt{3}$, then $\cos A \sin C + \sin A \cos C =$ (a) -1 (b) 0 (c) 1 (d) $\sqrt{3}/2$	1
9	Find the value of k for which the lines $-2x + 5y = 15$ and $-10x - ky = 75$ have infinite number of solutions. (a). 5 (b) -5 (c) 25 (d) -25	1
10	The distance between the points (a, b) and (-a, -b) is: (a). $2a$ (b) $2\sqrt{(a^2 + b^2)}$ (c) $2b$ (d) $2\sqrt{(a^2 - b^2)}$	1
11	An integer is chosen between 50 and 100. What is the probability that it is not divisible by 7? a) $42/51$ b) $42/49$ c) $43/50$ d) $44/51$	1
12	$\cos 60^\circ \sin 30^\circ + \sin 60^\circ \cos 30^\circ$ (a) -1 (b) 1 (c) 2 (d) 3	1

13	A card is randomly selected from a pack of 52 cards. What is the probability of getting neither a red card nor a black king a) $\frac{6}{13}$ b) $\frac{7}{13}$ c) $\frac{23}{26}$ d) none	1
14	A pole 6m high casts a shadow $2\sqrt{3}$ m long on the ground, then the Sun's elevation is (a) 60° (b) 45° (c) 30° (d) 90°	1
15	If $P(E) = 0.23$, what is the probability of 'not E'? a) 0.77 b) 0.023 c) 23 d) 0.0023	1
16	If the perimeter and the area of a circle are numerically equal, then the radius of the circle is: (a). 2 units (b) π units (c) 4 units (d) 7 units	1
17	In the given figure, if PR is tangent to the circle at P and Q is the centre of the circle and $\angle QPR = 60^\circ$ degrees, then $\angle POQ =$ 	1
18	If α, β are the zeroes of the polynomial $x^2 - 16$, then $\alpha\beta (\alpha + \beta)$ is: (a) 0 (b) 4 (c) -4 (d) 16	1
19.	Statement A (Assertion): $-5, -5/2, 0, 5/2, \dots$ is in Arithmetic Progression. Statement R (Reason) : The terms of an Arithmetic Progression cannot have both positive and negative rational numbers. (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A) (b) Both assertion (A) and reason (R) are true and reason (R) is not correct explanation of assertion (A) (c) Assertion (A) is true but reason (R) is false. (d) Assertion (A) is false but reason (R) is true.	1
20.	Statement A (Assertion): right circular Cylinder of radius 'r' and height 'h' ($h > 2r$) just encloses a sphere, then the diameter of sphere is '2r' Statement R (Reason) : The surface area of the sphere is $2\pi r(h+r)$. (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A) (b) Both assertion (A) and reason (R) are true and reason (R) is not correct explanation of assertion (A) (c) Assertion (A) is true but reason (R) is false. (d) Assertion (A) is false but reason (R) is true.	1
SECTION B [2 X 5 = 10]		
21.	A ladder, leaning against a wall, makes an angle of 60° with the horizontal. If the foot of the ladder is 2.5 m away from the wall, find the length of the ladder OR A circus artist is climbing a 20 m long rope, which is tightly stretched and tied from the top of a vertical pole to the ground. Find the height of the pole, if the angle made by the rope with the ground level is 30° .	2

22	A tangent PQ at a point P of a circle of radius 5 cm meets a line through the centre O at a point Q so that OQ = 12 cm. Calculate the length of PQ?	2																
23	Evaluate $2 \tan^2 45^\circ + 2 \cos^2 45^\circ - 2 \sin^2 45^\circ + \cot^2 45^\circ$ OR If $\cos A = 2/5$, find the value of $4 + 4 \tan^2 A$	2																
24	E and F are points on the sides PQ and PR, respectively of a ΔPQR . State whether $EF \parallel QR$ if $PQ = 1.28$ cm, $PR = 2.56$ cm, $PE = 0.18$ cm and $PF = 0.36$ cm	2																
25	Area of a sector of a circle of radius 36 cm is 54π cm ² . Find the length of the corresponding arc of the sector.	2																
SECTION C [3 X 6 =18]																		
26	Prove that: $(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$	3																
27	If we add 1 to the numerator and subtract 1 from the denominator, a fraction reduces to 1. It becomes $\frac{1}{2}$ if we only add 1 to the denominator. What is the fraction? OR For what value of k , $2x + 3y = 4$ and $(k + 2)x + 6y = 3k + 2$ will have infinitely many solutions.	3																
28	The length, breadth and height of a room are 8 m 25 cm, 6 m 75 cm and 4 m 50 cm respectively. Determine the length of the longest rod which can measure the three dimensions of the room exactly OR Prove that $\sqrt{7}$ is an irrational number	3																
29	A circle touches the side QR of a ΔPQR at 'M' and side PQ and PR on producing at 'S' and 'T' respectively. If PS=8cm. Find the perimeter of ΔPQR . 	3																
30	Compute the mode for the following frequency distribution. <table border="1" data-bbox="303 1402 1357 1482"><tr><td>Size of items(in cm)</td><td>0-4</td><td>4-8</td><td>8-12</td><td>12-16</td><td>16-20</td><td>20-24</td><td>24-28</td></tr><tr><td>frequency</td><td>5</td><td>7</td><td>9</td><td>17</td><td>12</td><td>10</td><td>6</td></tr></table>	Size of items(in cm)	0-4	4-8	8-12	12-16	16-20	20-24	24-28	frequency	5	7	9	17	12	10	6	3
Size of items(in cm)	0-4	4-8	8-12	12-16	16-20	20-24	24-28											
frequency	5	7	9	17	12	10	6											
31	If one zero of the polynomial $(a^2 + 9) x^2 + 13x + 6a$ is reciprocal of the other. Find the value of a.	3																
SECTION D [5 X 4 = 20]																		
32	If the median of the distribution given below is 28.5, find the values of x and y. <table border="1" data-bbox="303 1631 1250 1808"><tr><td>Class interval</td><td>0-10</td><td>10-20</td><td>20-30</td><td>30-40</td><td>40-50</td><td>50-60</td><td>total</td></tr><tr><td>frequency</td><td>5</td><td>x</td><td>20</td><td>15</td><td>y</td><td>5</td><td>60</td></tr></table>	Class interval	0-10	10-20	20-30	30-40	40-50	50-60	total	frequency	5	x	20	15	y	5	60	5
Class interval	0-10	10-20	20-30	30-40	40-50	50-60	total											
frequency	5	x	20	15	y	5	60											
33	State and prove basic proportionality theorem. OR	5																

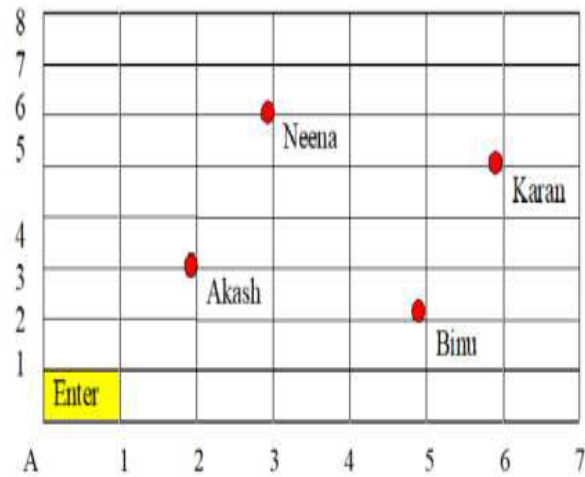
	Sides AB and AC and median AD of a triangle ABC are respectively proportional to sides PQ and PR and median PM of another triangle PQR. Show that $\Delta ABC \sim \Delta PQR$.	
34	The age of a man is twice the square of the age of his son. Eight years hence, the age of the man will be 4 years more than three times the age of his son. Find their present ages.	5
35	A vessel is in the form of an inverted cone. Its height is 8 cm and the radius of its top, which is open, is 5 cm. It is filled with water up to the brim. When lead shots, each of which is a sphere of radius 0.5 cm, are dropped into the vessel, one-fourth of the water flows out. Find the number of lead shots dropped in the vessel. OR Suhana's house has an overhead tank in the shape of a cylinder. This is filled by pumping water from a sump (an underground tank) which is in the shape of a cuboid. The sump has dimensions $1.57 \text{ m} \times 1.44 \text{ m} \times 95 \text{ cm}$. The overhead tank has a radius of 60 cm and a height of 95 cm. Find the height of the water left in the sump after the overhead tank has been completely filled with water from the sump which had been full. Compare the capacity of the tank with that of the sump. (Use $\pi = 3.14$)	5
SECTION E [4 X 3 =12] CASE STUDY BASED QUESTIONS		
36.	A group of students of class X visited India Gate on an education trip. The teacher and students had interest in history as well. The teacher narrated that India Gate, official name Delhi Memorial, originally called All-India War Memorial, monumental sandstone arch in New Delhi, dedicated to the troops of British India who died in wars fought between 1914 and 1919. The teacher also said that India Gate, which is located at the eastern end of the Rajpath (formerly called the Kingsway), is about 138 feet (42 metres) in height.  1. What is the angle of elevation if they are standing at a distance of 42m away from the monument? a) 30° b) 45° c) 60° d) 0°	1+1+2

	2. They want to see the tower at an angle of 60°. So, they want to know the distance where they should stand and hence find the distance. a) 25.24 m b) 20.12 m c) 42 m d) 24.64 m 3. The ratio of the length of a rod and its shadow is 1:1. The angle of elevation of the Sun is a) 30° b) 45° c) 60° d) 90° OR If the altitude of the Sun is at 60°, then the height of the vertical tower that will cast a shadow of length 20 m is a) $20\sqrt{3}$ m b) $20/\sqrt{3}$ m c) $15/\sqrt{3}$ m d) $15\sqrt{3}$ m	
37.	India is competitive manufacturing location due to the low cost of manpower and strong technical and engineering capabilities contributing to higher quality production runs. The production of TV sets in a factory increases uniformly by a fixed number every year. It produced 16000 sets in 6th year and 22600 in 9th year. Based on the above information, answer the following questions:  (i) Find the production during first year. (ii). Find the production during 8th year. (iii). Find the production during first 3 years. OR In which year, the production is Rs 29,200	1+1+2

38

Karan went to the Lab near to his home for COVID 19 test along with his family members

1+1+2



The seats in the waiting area were as per the norms of distancing during this pandemic (as shown in the above figure). His family members took their seats surrounded by a red circular area.

- (i) What is the distance between Neena and Karan?
- (ii) What are the coordinates of the seat of Akash?
- (iii) What will be the coordinates of a point exactly between Akash and Binu where a person can be seated?

OR

Find the coordinates of mid-point of Neena and Binu.